GENRE-AWARE AUTOMATIC VOCAL MELODY HARMONIZATION USING HIERARCHICAL DEEP LEARNING

Vocal Input Validation and Minimal Musical Repair using Boundary-Pitch Disentangled Learning

J26-DS-339

Project Proposal Report

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September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Novice singers can still express a useful melody even when their recordings contain small pitch errors, noise, or unclear changes between notes. Before using this type of recording for automatic harmonization, it needs to be handled carefully. If the system makes a mistake when identifying a note or its boundary, correcting that mistake should not change a phrase that was already sung correctly. Previous research on context aware pitch correction depends on note segmentation [1], while studies using solo vocal annotations show that note boundaries are not always clear [2].

The research purpose Boundary Pitch Disentangled Repair (BPDR) for the input stage of a genre aware vocal melody harmonization system. The main research question is whether the model can produce stable pitch corrections when note boundaries are incorrectly split or merged, while still responding correctly to small pitch errors in the singing. The purposed contribution is a paired training method that compares correction curves over the same time positions and includes targets that help preserve parts of the performance that should not be changed. Combining audio libraries, rules and a transformer is not the claimed novelty.

The study will use controlled changes to audio, a small public vocal dataset, and recordings collected from participants after ethics approval. A compact transformer model will allow only limited changes that are supported by the available evidence. Recording that are already usable will be kept musically unchanged, while unsuitable recordings will receive a clear request to record again. The prototype will be developed step by step combining acoustic checks, careful audio preparation, local pitch correction, and a traceable output record.

Keywords: vocal input validation; minimal pitch repair; note segmentation; melody harmonization.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude for our supervisor Prof. Dilshan De silva, for his continuous guidance, advice and feedback given throughout the research. His expertise and guidance have been invaluable in shaping the direction of this study.

I would also like to appreciation for our co-supervisor, Prof. Samantha Thelijjagoda, for his suggestions, technical guidance, and support given throughout the development of this proposal.

This research is conducted as collaborative undergraduate research project. Although this document identifies my individual research component, I would like to acknowledge the cooperation, discussions and teamwork of my fellow group members:

- Perera P D (IT23148086)
- Perera E V (IT23439078)
- Jayakody J A S R (IT23157132)

Their collaboration and our team's shared commitment contributed into shaping and planning the overall vision of the complete research project.

Finally, I would like to thank the academic staff of the Sri Lanka Institute of Information Technology (SLIIT), as well as my family and friends for their continuous encouragement, and unwavering support through the preparation of this research proposal

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
APC	Automatic pitch correction
BPDR	Boundary-Pitch Disentangled Repair
DPW	Dynamic Pitch Warping
F0	Fundamental frequency
GPU	Graphics processing unit
JSON	JavaScript Object Notation
MIR	Music information retrieval
SDG	Sustainable Development Goal
STFT	Short-time Fourier transform
WBS	Work breakdown structure

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Appendix 1 - Proposed Research Instruments and Experimental Records

Appendix 2 - Mandatory Proposal Evidence and Decision Log

Appendix 3 - AI Use Disclosure

1. INTRODUCTION

1.1 Background and Context

A user should be able to communicate a melody by singing or humming without needing to provide note names, a key signature, or written music. In this project, the recording will be used as the input for melody extraction and later chord generation. Therefore, the first component must decide whether the recording contains enough reliable musical information, whether a small amount of preparation or correction would improve it, or whether asking the user to record the phrase again would be more suitable.

Recording quality and musical usability are not the same. For example, reducing steady background noise may make the voice clearer, but it cannot determine the melody that the user intended. In the same way, accurately measuring the pitch being sung does not prove that the note was intentional. In this study, a recording will be considered musically usable when it provides suitable input for the agreed melody extractor while preserving the order, timing, and expressive variation of the performed phrase. The system does not require professional singing or perfect matching to an assumed musical scale.

The available evidence supports a focused study rather than a claim about the wider population. The vocadito dataset contains 40 solo-vocal recordings in seven languages and includes two independent sets of note annotations [2]. These figures describe the dataset only and do not show how common singing errors are among Sri Lankan users. Therefore, this study will not assume a local error rate. Instead, the planned participant recordings will be used to document the types of errors that actually occur in the target recording workflow.

The first version of the system will accept short, unaccompanied recordings containing only one voice. A maximum length of 20 seconds is proposed to keep processing and annotation manageable. Suitable recordings may go through technical checks, optional reduction of steady background noise, and small pitch-center corrections. The system will preserve the original rhythm, syllable timing, and rests. It will not create a melody from normal spoken text, recover heavily clipped recordings, or separate multiple overlapping singers. Chant-like singing may still be accepted, and speech-like delivery by itself will not automatically cause the recording to be rejected.

This research is mainly in the area of artificial intelligence and machine learning, with music information retrieval as the main application area. This only shows how the project fits thematically within these fields and does not mean that it has received formal approval under a specific research cluster. The study also relates to SDG 9 because it supports research and innovation [3]. However, any wider benefit to creative participation would need to be supported by user evidence. The prototype alone will not be used to claim national economic or educational impact.

1.2 Literature Review

Context-aware and continuous correction

Kim et al.'s BERT-APC uses note segmentation, stationary-pitch estimation, and MusicBERT-based musical context before applying pitch correction to the audio [1]. The study uses AI-Hub singing datasets together with an in-house collection. Its results show that the method can handle small errors in note-boundary positions, although performance decreases as the boundary errors become larger. However, the boundary experiment only shifts existing boundaries and does not test false note splits or incorrectly merged notes. This makes BERT-APC an important comparison for the proposed research. However, its reported results cannot be directly applied to novice free humming because the data and recording conditions are different.

Wager et al. proposed a continuous pitch-correction method for karaoke recordings using separate but aligned vocal and accompaniment tracks [4]. Their training data was based on semi-professional singing. A main strength of the method is that it produces smooth, context-aware corrections instead of forcing every note to a fixed pitch grid. However, it depends on accompaniment information, while the proposed BPDR system is designed for unaccompanied singing or humming. This work prevents claiming continuous correction or vibrato preservation as new contributions.

Molina Villota et al. propose Dynamic Pitch Warping for expressive vocal retuning [5]. Their signal-processing approach uses pitch movement and a target scale to retain ornamentation. The evaluation concerns expressive retuning examples, not a novice-input acceptance study. It provides a useful preservation principle, but scale-directed tuning does not establish the correction intended for an ambiguous free-hummed phrase.

Schwär and Müller proposed a method for improving the tuning of multiple voices without using note-level information [6]. Their results show that pitch correction can be performed without depending on note segmentation. However, their work focuses on several voices at the same time, while this study focuses on one vocal recording. Even so, their approach suggests an important alternative: instead of trying to make boundary information more reliable, it may be simpler to remove boundary information completely. This is why the proposed study includes a matched frame-only Transformer as a comparison.

Pitch analysis, transcription and learning principles

SPICE learns pitch relationships from transformed spectral inputs and evaluates pitch estimation on vocal datasets including MIR-1K [7]. PESTO uses a transposition-equivariant objective and evaluates music and speech material, including MIR-1K, MDB-stem-synth and PTDB [8]. Both show that transformation-based learning is established. PESTO's available implementation makes it a practical frozen analyzer, but an estimate of performed pitch is not a target melody or permission to edit.

ROSVOT uses multi-scale context and noise-robust training for singing transcription, with experiments on singing material including M4Singer [9]. Its synthesis-oriented evaluation addresses the usefulness of transcribed notes. Transcription robustness is valuable, but it does

not by itself demonstrate that modifying the original waveform is warranted. This distinction requires separate extraction and preservation measurements in the proposed study.

Mean Teacher uses consistency targets in semi-supervised image-learning experiments [10]. Its domain differs, but its methodological relevance is direct: consistency regularization is not a new general principle. BPDR must be assessed as a particular repair-response formulation, and against identical augmented training data without the additional response terms.

Existing products and their relevance

Melodyne provides interactive pitch-centre, modulation and drift controls, as well as timing and note-separation editing [11], [12]. Its documented melodic processing distinguishes pitched and unpitched material. These capabilities for musicians and audio editors extend beyond nearest-note correction. The cited product pages do not provide a controlled performance benchmark for this project's input-acceptance task. The proposed component will address automated, limited preparation and an explicit hand-off decision, rather than attempt to replace a full vocal editor.

Table 1. Literature implications for the proposed study

Evidence	Capability already established	Consequence for this proposal
BERT-APC [1]	Reference-free contextual correction	Test split/merge changes separately from jitter.
Wager et al. and DPW [4], [5]	Continuous, expressive retuning	Measure preservation; do not claim it as an invention.
Multi-voice adaptation [6]	Correction without note-level input	Include a matched frame-only model.
SPICE, PESTO and Mean Teacher [7], [8], [10]	Transformation and consistency learning	Compare the same data with and without response losses.
ROSVOT and vocadito [9], [2]	Transcription methods and annotation evidence	Use independent references and downstream tests.

1.3 Research Gap

The reviewed studies do not establish whether a minimal vocal-repair model can remain stable under plausible split/merge corruption of provisional boundaries while preserving a localized corrective response to genuine small detuning. the research gap is whether these two behaviors can be learned separately during input preparation: ignoring unreliable boundary changes while still making suitable local pitch corrections.

A false split changes how the system represents the note boundaries, but the actual audio remains unchanged. In contrast, a small detuning intervention changes the pitch of the audio while keeping the intended note the same within the controlled experiment. BPDR will represent all corrections on the same frame-by-frame time grid and will learn different responses to these two types of changes. This avoids the need to match note numbers between versions that may contain different numbers of detected notes.

The proposed novelty is this specific boundary-invariance and local counter-detuning objective with supervised preservation anchors. Neither the Transformer nor the integration of existing tools is claimed as novel. The hypothesis will be supported only if the additional objective improves held-out repair and preservation relative to augmentation-only and frame only controls. An outcome explained entirely by those simpler approaches will not support the proposed methodological advantage. The proposal makes no worldwide priority claim.

2. OBJECTIVE

2.1 Main Objective

To develop and experimentally evaluate BPDR for short vocal-input preparation, determining whether it improves local pitch repair and subsequent melody extraction under segmentation errors without increasing harmful alteration of acceptable musical material at matched acceptance coverage.

2.2 Specific Objectives

1. Characterize false splits, merges, pitch deviations and recording failures in the supported input workflow, producing an annotated failure taxonomy and baseline measurements.

2. Construct a reproducible dataset pipeline with accepted reference phrases, controlled audio and boundary interventions, documented permissions and source-separated experimental partitions.

3. Implement the BPDR objective and compact correction model and compare it with an augmentation-only Transformer and a capacity-matched frame-only Transformer using the same source partitions, training data, optimization budgets and evaluation conditions.

4. Develop the input component with technical checks, bounded waveform preparation, unchanged-input acceptance, reasoned re-recording requests and a traceable hand-off package.

5. Evaluate repair accuracy, extraction benefit, expression preservation, decision quality and computational cost using held-out recordings, independent annotations and blinded listening.

3. METHODOLOGY

3.1 Research Design

The research will use controlled comparison experiments together with step-by-step prototype development. The main aim is to test whether the proposed training objective provides an improvement, so the study will compare matched models under controlled conditions instead of only building a working application. Controlled data will be used to separate the effects of boundary errors and small pitch deviations. Recordings from novice singers will then be used to check whether the results also apply to more natural input. Listening tests will be included because pitch-based measurements alone cannot fully describe musical quality or audio rendering problems.

The main hypothesis is that BPDR will reduce pitch correction errors and remain more stable when note boundaries are incorrectly split or merged. This will be tested against an augmentation-only Transformer trained using the same four-view data but without the BPDR response losses. A second comparison will use a similar frame-only Transformer to check whether simply removing boundary information is a better solution. Original audio and conservative rule based correction will provide non-learning reference conditions. The study will also test whether audio prepared by BPDR improves the output of a fixed melody extraction model. These two parts will be evaluated separately because better pitch correction may not always lead to better melody extraction when the extractor can already handle small pitch errors.

3.2 Data Participants and Experimental Inputs

The vocadito dataset will be used as a small public dataset for development and for studying note-boundary annotations [2]. Its note labels will not be treated as completely reliable representations of the singer's intended notes. The main study will also collect vocal recordings from consenting adult participants after institutional ethics approval. These recordings will include accepted reference phrases, short guide-melody performances, and free humming. Guide-melody information will be used only for evaluation and will not be given to the model during normal system use.

A proposed pilot will involve eight adults producing six short phrases each, giving up to 48 source recordings. These numbers are only planning estimates and do not represent completed recruitment or a formal sample-size calculation. The pilot will be used to estimate how many recordings are usable, annotation time, how much variation appears between singers, and whether accepted reference recordings can be identified. The final sample size will be decided before the main test data is collected, based on the pilot results, available resources, and a stated precision or power calculation. The study will also record exclusions and cases where participants do not complete the study. If enough independent data cannot be collected, the findings will be reported as a pilot study rather than making broader claims based mainly on augmented data.

Training targets will start from performances judged suitable to preserve. Two reviewers with relevant musical experience will check whether the reference recordings are suitable and

whether any note transitions are unclear. If they disagree, the disagreement will be recorded and resolved with a stated reason where needed. The study will use original or properly permitted melodic material. Singer and phrase identities will be stored using coded labels, while personal contact details will be kept separate from the audio recordings.

Singers and their original recordings will be divided into training, development, and test sets before any augmented versions are created. All modified versions of the same source recording will stay in the same set to avoid data leakage. A separate test using unseen melodies will be used when making claims about unfamiliar tunes. Part of the development set will be used to choose controller thresholds. The final test set will not be used to select model parameters, loss weights, or decision thresholds.

Four views will be constructed from each accepted source. Views A and B will use the same audio, but A will use the normal boundary information while B will use intentionally changed boundary information. Views C and D will use the same locally detuned audio, again with normal and changed boundary information. The boundary changes will include false splits, missing boundaries, and small timing shifts. The pitch experiments will initially use changes of +10, +20, +30, -10, -20, and -30 cents within stable parts of notes, with smooth transitions into and out of the changed region. These values are experimental settings used to test the model and should not be treated as final correction limits.

After changing the pitch, the audio will be rendered again and all acoustic features will be recalculated. Identity-rendered controls will be used to check whether the rendering process itself introduces noticeable changes. Any intervention will be rejected if the actual pitch shift does not match the intended target closely enough. No-change test cases will include vibrato, slides, and acceptable whole-phrase transpositions to check whether the system avoids unnecessary correction. Natural analyzer errors will also be tested as a separate scenario using the same five evaluation conditions. Artificially modified recordings will not be treated as recordings from new singers or used as complete evidence of real-world performance.

3.3 Baselines and Evaluation Conditions

Table 2 presents the five evaluation conditions used in this study. Only three of these require neural model training: the augmentation-only Transformer, the matched frame-only Transformer, and the full BPDR Transformer. These models will use the same data partitions, the same four-view training data, similar training targets, comparable model capacity, and matched training and evaluation settings. The augmentation-only model will be the main baseline for testing whether the BPDR response objective provides an additional benefit. The frame-only model will test whether simply removing boundary information is enough. Original audio and conservative rule-based correction will be used as non-learning reference conditions. Systems discussed in the literature review will be used only for background comparison and will not be included as additional experimental models.

Table 2. Evaluation conditions and comparison purpose

Evaluation condition	Type	Purpose
Original audio	Non-learning reference	Establish whether preparation improves the original input at all.
Conservative rule-based correction	Non-learning reference	Test whether simple bounded correction is sufficient without learned repair.
Augmentation-only Transformer	Trainable model	Main BPDR baseline: identical four-view data without boundary and detuning response losses.
Matched frame-only Transformer	Trainable model	Test whether removing provisional boundary information is sufficient at comparable capacity.
Full BPDR Transformer	Trainable proposed model	Test the complete boundary-invariance and local counter-detuning training objective.

The rule-based method will apply limited chromatic pitch-centre correction after estimating the overall tuning offset, with the same rendering and permission limits wherever possible. The downstream melody extractor will remain unchanged so that results before and after preparation can be compared fairly. All trainable models will use the same source partitions and the same evaluation conditions.

Technical preparation will be used as a shared processing step for the correction methods and will not be treated as a separate evaluation condition. The original-audio reference will only receive the decoding and format conversion required by the fixed melody extractor, without noise reduction or pitch correction. If noise handling is used in the correction methods, the same fixed policy will be applied to all of them. The evaluation will test the complete BPDR objective as one method and will not try to measure the individual contribution of each response-loss term separately.

3.4 Evaluation Metrics

The main evaluation measure will be the absolute pitch correction error in cents, measured on controlled and reliable voiced parts of the recording. The main comparison will be between the full BPDR model and the augmentation-only Transformer because both use the same four-view training data, while only BPDR includes the proposed response objective. A second comparison will be made with the matched frame-only Transformer to check whether using boundary information gives an advantage over removing it. Original audio and conservative rule-based correction will be used as practical reference conditions. For recordings with a known added pitch error, the expected correction will be the opposite of the added shift. The pitch of the rendered audio will also be compared with the accepted reference, because predicting the correct adjustment does not always mean that the final audio was corrected properly. Melody extraction will be evaluated using note precision,

recall, and F1 against independent references. The mir_eval transcription method and its defined matching rules will be used [13], with onset-and-pitch and offset-aware results reported separately.

Table 3. Evaluation metrics and comparison criteria

Metric	Why relevant	Target or comparison
Correction error in cents	Measures whether local repair matches the known compensating correction	Primary: lower paired BPDR error than augmentation-only under held-out interventions. Also report frame-only, rule based and original-audio references.
Note precision, recall and F1	Tests benefit to the next component	Compare fixed-extractor results before and after preparation.
Boundary sensitivity	Measures prediction change when only split/merge metadata changes	BPDR should be less sensitive than augmentation-only. The frame-only model provides a boundary-free reference.
Edit magnitude and coverage	Quantifies unnecessary intervention	Report on accepted no-change phrases and alongside repair gains.
Harmful-edit rate and acceptance coverage	Prevents success by changing too much or rejecting everything	Compare at matched coverage; report uncertainty and abstentions.
Listening judgements	Assesses melodic identity, expression and artefacts	Blinded paired judgements with disagreement reported.
Latency, memory and render failures	Tests deployability	Benchmark short phrases on the documented demonstration hardware.

Boundary sensitivity will be measured for models that use boundary information by giving them the same audio with normal and intentionally changed boundary metadata. The frame-only model should produce the same output in both cases because it does not receive boundary information. This will be used only as a reference check and will not show whether the model performs good pitch repair. Original audio and other non-learning reference conditions that do not use boundary metadata will be marked as not applicable for this measure.

An edit will be considered harmful if independent reference checks or the planned listening test show that it creates a new melody error or causes an unacceptable loss of expression or audio quality. Listeners will judge tune preservation and rendering quality separately. The listening criteria, decision thresholds, and acceptable preservation margin will be decided using the pilot study before the final evaluation begins. The study will not assume that there is one universal score that can determine whether a musical result is valid.

3.5 Validation Strategy

The study will report paired differences with 95% confidence intervals. Singers will be treated as the main independent unit, or individual recordings will be used when singer identity is not available. Frames from the same recording will not be treated as separate participants. Where computing resources allow, the trainable models will be tested using three preselected random seeds so that training variation can be reported separately from data uncertainty. The main learned comparison will be between the full BPDR model and the augmentation-only Transformer, using held-out correction error under controlled split/merge and local-detuning conditions. The frame-only Transformer will be used to test whether removing boundary information is a simpler alternative. Original audio and rule-based correction will be treated as reference conditions rather than additional neural-model hypotheses.

The same number of training runs, training budget, and model-selection process will be used for all three trainable models. If computing limits require fewer training runs, this decision will be recorded before the final training and before the test data is used. If only one training run is possible, the result will be reported with this limitation and will not be treated as evidence of training stability. Other evaluation measures will be used to help explain and support the main comparison.

Results will be examined separately for different types of boundary errors, levels of pitch deviation, expressive singing features, and recording difficulty. The same audio will be tested with normal and changed boundary information to measure how sensitive each boundary-based model is to boundary errors. Because the location of the added pitch change is known, the study will also measure whether the model makes unnecessary corrections outside that region. Natural analyser errors and identity-rendered controls will be included to check that the model has not learned only from artificial changes. A low boundary-sensitivity score alone will not be treated as a successful result if the model simply produces no correction or predicts the wrong melody.

Listening tests will be carried out without showing participants which method produced each audio sample, and the order of the samples will be randomized. For free humming, where there may be no exact written reference melody, the singer's own judgement of whether the musical idea was preserved will be considered together with independent listener assessments. Agreement and disagreement between listeners, unclear examples, and cases that require another recording will also be reported. Any claimed improvement will need to show both better pitch correction and acceptable preservation of the original musical idea.

3.6 Development Approach

The system will be developed step by step because feature alignment, audio rendering, and the connection with the downstream melody extractor need to be tested together. The first stage of development will establish the unchanged-audio path, diagnostic outputs, and failure handling. The next stage will create and verify the four-view training data. The augmentation-only Transformer, matched frame-only Transformer, and full BPDR Transformer will then be implemented and trained under matched conditions. BPDR

correction, bounded audio rendering, and user feedback will be added before final integration. The final system will use fixed configurations and a held-out test set for evaluation.

The initial model will use a four-layer bidirectional Transformer encoder implemented in PyTorch [14]. It will use a 128-dimensional representation, four attention heads, a 512-dimensional feed-forward layer, and a dropout rate of 0.1. These are starting settings and may be adjusted using development data. With a frame step of approximately 20 ms, a 20-second recording will contain about 1,000 frames. The system is therefore designed mainly for offline processing of short vocal phrases. Available computing resources will determine practical batch sizes and deployment settings.

The augmentation-only and full BPDR models will use 71 input features. These include 64 log-mel features, performed pitch, pitch change between neighbouring frames, pitch confidence, a voiced mask, log energy, onset strength, and a provisional boundary score. Training will also use normalization, positional information, and padding masks. Missing pitch values will be represented using a mask instead of being treated as zero-Hz notes. The model will predict a relative pitch correction in cents. If a proposed correction is too large, the controller will reject it instead of simply clipping the value.

The frame-only model will use the same features except for the provisional boundary score, giving it 70 input features. It will use the same Transformer depth and hidden dimensions as the other models. Parameter counts and training budgets will be recorded to make sure the models have comparable capacity. Its four training views will use the same source weighting and acoustic changes. However, because the frame-only model does not receive boundary information, the nominal and corrupted-boundary versions of the same audio will have identical inputs.

Let rA, rB, rC and rD be the four predicted correction curves, e the known local detuning, and dM the mean absolute difference over a common reliable voiced-frame mask M. The proposed response terms are:

$$L_B = \frac{1}{2}[d_M(r_A, r_B) + d_M(r_C, r_D)]$$

$$L_D = \frac{1}{2}[d_M(r_C - r_A, -e) + d_M(r_D - r_B, -e)]$$

The first term discourages different repairs when only boundary metadata changes. The second requires the repair to change by the negative of the added detuning, with no relative change outside its support. This relation is limited to small interventions that preserve the intended note. It will not be applied as a rule for arbitrary transposition.

A supervised anchor loss L_A will compare predictions with trustworthy correction targets: zero for accepted originals and the compensating target for verified detuned views. L_S will penalize abrupt changes in the correction curve within valid voiced spans, without flattening the original pitch contour. The full BPDR model will minimize:

$$L = L_A + \lambda_B L_B + \lambda_D L_D + \lambda_S L_S$$

The augmentation-only Transformer will use the same supervised anchor and smoothness losses but will not use the two BPDR response losses:

$$L = L_A + \lambda_S L_S$$

The matched frame-only Transformer will use the same anchor-and-smoothness objective, but its inputs will not contain boundary information. This provides a simpler learned comparison using the same acoustic data and correction targets. The supervised anchor loss and supervised warm-up will remain part of the training process for all three retained trainable models.

The loss weights will be selected using the development data. Cases where the voiced-frame mask is empty will be excluded from the loss calculation and recorded. Training will begin with a supervised warm-up and will then use the AdamW optimizer for the full objective. If the model could learn the supervised targets perfectly, the desired response relationships would already be satisfied. Therefore, the additional BPDR response losses are intended to test whether they improve learning and generalization when the available data is limited. The augmentation-only Transformer is important for this test because it is trained on the same data but without the BPDR response losses.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of Proposed Solution and Its Components

During normal use, the input component will include one fixed neural pitch estimator, the trained BPDR Transformer, and a set of deterministic signal-processing and permission rules. The augmentation-only and frame-only Transformers will be used only for offline evaluation and will not be placed one after another in the final system. Therefore, the proposed method is a hybrid system that combines neural models with fixed processing rules. The rules will check file validity, control the allowed amount of correction, and decide the final hand-off status. They will not attempt to represent all musical knowledge or assume the user's intended melody with complete certainty. Python will control the overall processing pipeline, while Matplotlib will be used to create plots from the recorded evaluation results.

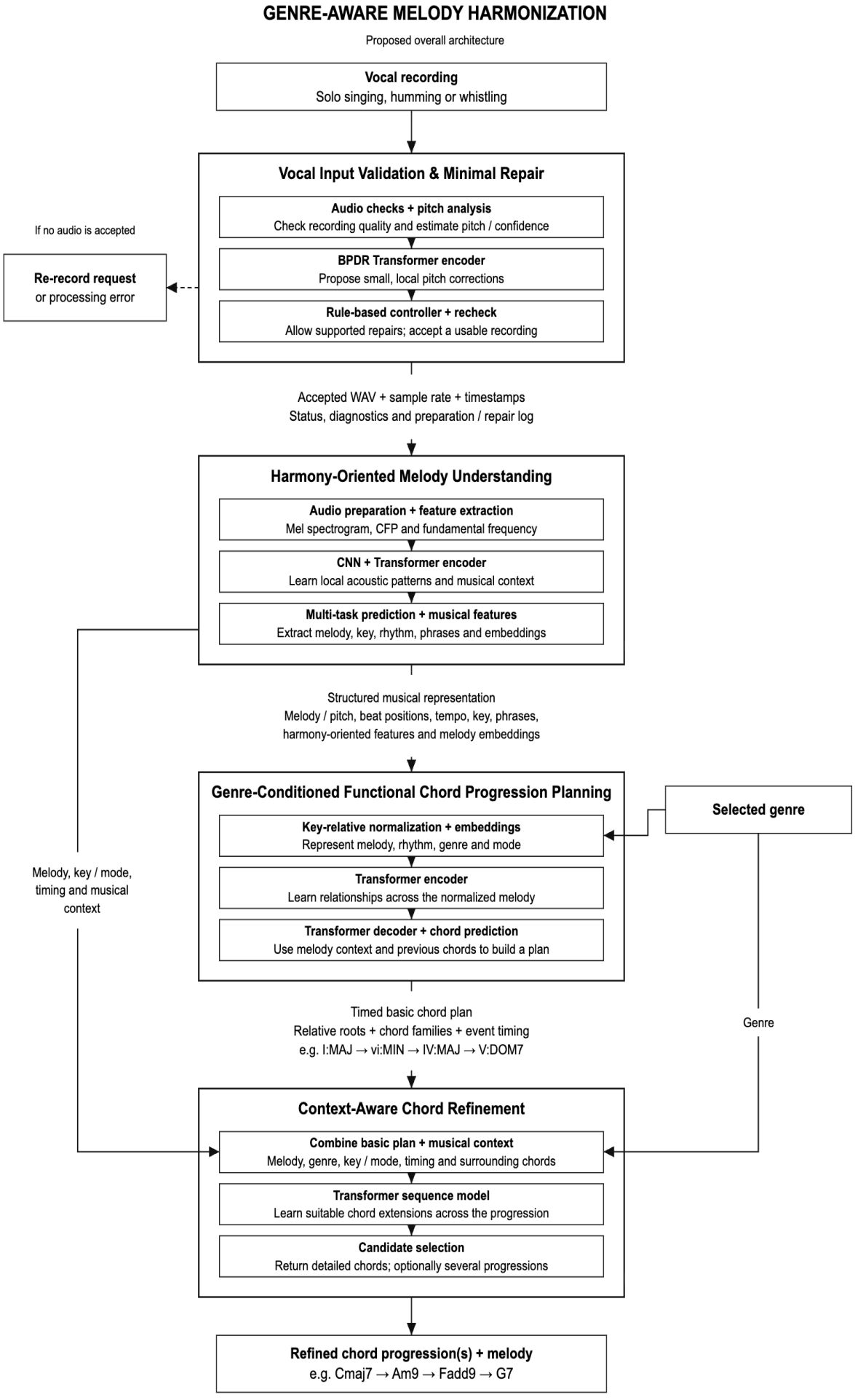


Figure 1. Overall system architecture and responsibility boundary

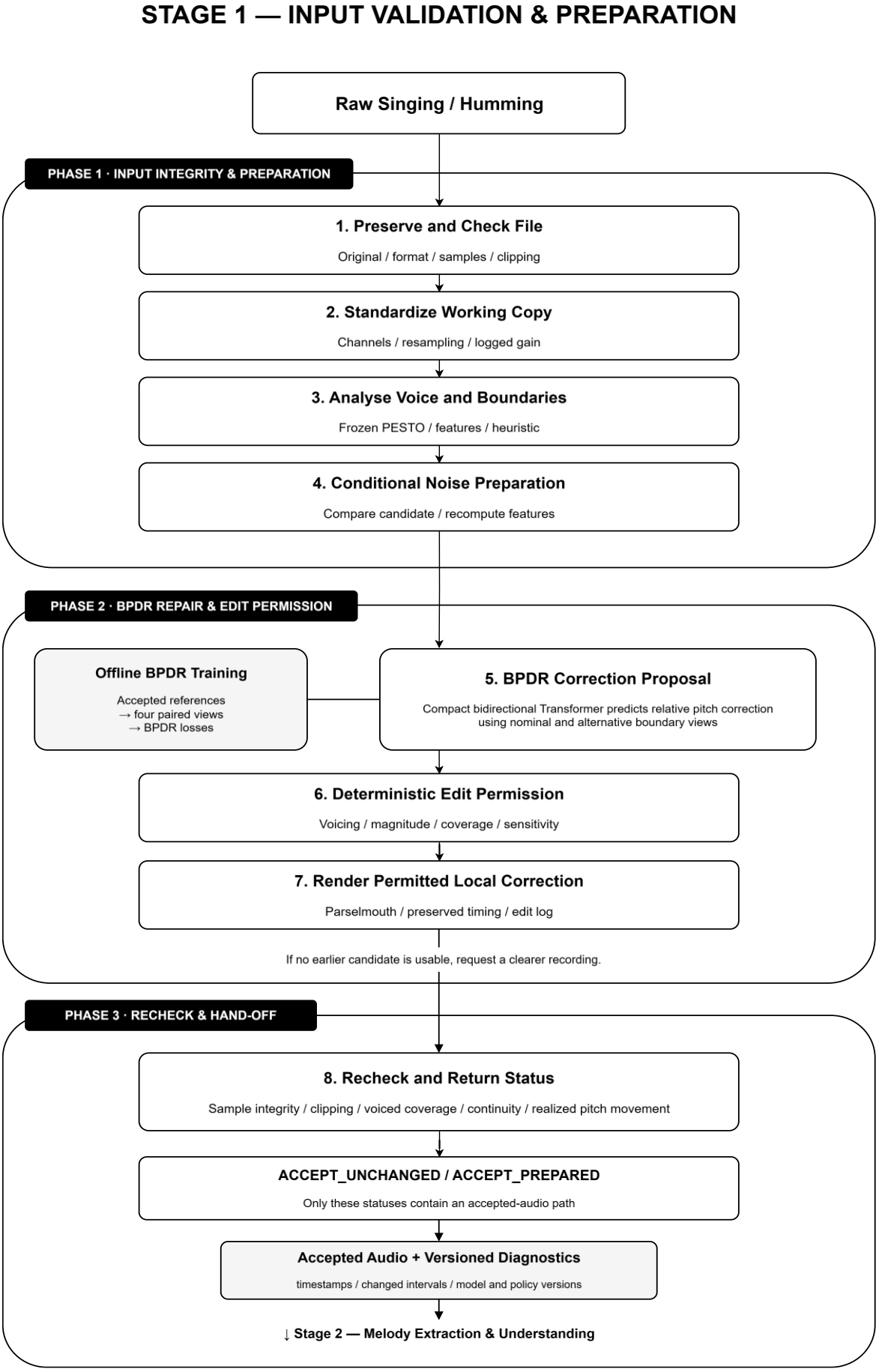


Figure 2. Component workflow

Processing will begin by keeping the original audio file and recording its checksum so that the input can be traced later. SoundFile will be used to decode WAV or FLAC files [15]. NumPy checks will identify empty or invalid audio data and possible clipping. For stereo recordings, the system will first examine the energy and similarity between the two channels before converting them to mono. This avoids simply averaging the channels when doing so could reduce or cancel the vocal signal. SciPy polyphase resampling will then create a mono working copy at a proposed sampling rate of 22,050 Hz [16]. The original timing of the recording will be preserved, and any overall change in audio gain will be recorded.

The system will use the frozen PESTO model to estimate the performed pitch and its confidence, starting with the documented mir-1k_g7 checkpoint [17]. Librosa will be used to extract spectral features and onset strength [18]. A fixed heuristic will then combine onset information, lasting pitch changes, and voiced/unvoiced transitions to produce provisional boundary scores. These boundary scores will be treated as uncertain input features and not as final note labels.

Noise reduction will only be used when the recording contains a reliable section with background noise but no singing. In that case, a limited and smoothed spectral mask will be applied while keeping the original phase of the audio during reconstruction. Low pitch confidence by itself will not be treated as evidence of noise because breaths and consonants may naturally have weak or unclear pitch. If the noise-reduction step damages useful vocal information, that processed version will be rejected. After the final working waveform is selected, all acoustic features will be calculated again before the audio is passed to the BPDR model.

The model's main correction prediction will be compared with predictions produced using up to two other reasonable boundary versions. If the predicted correction changes too much between these boundary versions, or if the system detects uncertain voicing, protected note transitions, very large pitch changes, or too much of the recording being altered, that edit will be blocked. The decision thresholds will be selected together using the development data. The system will not force unfamiliar notes into a fixed key or scale. If a proposed correction is rejected, the original recording may still be accepted if it is already usable.

Approved local adjustments will be rendered using Praat through Parselmouth [19]. For voiced frame t, the target frequency will be the input frequency multiplied by 2 raised to r(t)/1200. Duration will not be changed. Local crossfades and their support will be logged. Re-analysis will check sample integrity, clipping, voiced coverage, continuity and realized pitch movement. A failed render will be rolled back to a usable earlier candidate or lead to a re-recording request.

Overall Integration

Only accepted audio will pass to the melody-extraction component. The package will contain the original identifier, accepted WAV, sample rate, timestamps, diagnostics, model and

policy versions, changed intervals and reasons. The four statuses will be ACCEPT_UNCHANGED, ACCEPT_PREPARED, RERECORD_REQUIRED and TECHNICAL_FAILURE. The last two will contain no accepted-audio path. Musical acceptance without repair may still involve documented format conversion.

The second component will own final melody extraction. Its output will feed basic chord generation, followed by harmonic refinement. Genre will belong to the project-level context and later musical decisions; it will not authorize melody replacement in this component. Integration will use a versioned Python interface and JSON metadata with seconds-based timestamps. Contract tests will prevent downstream fallback to rejected audio. Post generation harmony validation is outside this proposal.

Individual Responsibility Boundary

Table 4. Individual and shared responsibilities

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Dataset preparation		All Members	
Audio input quality checks	✓		
Pitch and boundary feature extraction	✓		
BPDR model development and training	✓		
Pitch correction safety checks	✓		
Validation feedback and diagnostic logs	✓		
Input validation component evaluation	✓		
Final melody extraction			Perera E V
Genre-conditioned harmonic planner			Perera P D
Detailed chord realization			Jayakody J A S R
Complete system evaluation		All Members	

5. USER REQUIREMENTS

The initial requirements are derived from the research owner's stated workflow and the limitations examined in Section 1. They are proposed requirements, not findings from completed interviews or surveys. Before implementation is fixed, novice users and the extraction owner will review recording instructions, acceptance messages and the output contract. Appendix 1 gives the planned elicitation and evaluation instruments. Changes will be recorded with their evidence and rationale.

Functional Requirements

FR1. The user shall record or upload a supported short vocal phrase, with understandable instructions on one voice, limited duration and avoiding accompaniment. Streamlit will provide capture, upload and playback in the local prototype [20].

FR2. The component shall retain the original and distinguish technical processing failures from requests for a clearer performance.

FR3. The component shall assess recording integrity and provisional melodic evidence, and either preserve usable input or prepare a separate candidate using permitted operations.

FR4. The component shall restrict pitch edits to supported local intervals and preserve timing, rests and event order. It shall not require note-name or key entry from the user.

FR5. The user shall be able to compare original and prepared playback, inspect changed intervals and choose to supply another take. A rejection message shall identify an observation and a practical recording action, without presenting uncertain musical inference as fact.

FR6. The component shall export accepted audio and versioned diagnostics only after permission and rendering checks. Failed candidates shall remain excluded from downstream processing.

FR7. The evaluation interface shall show aligned measured pitch plots and operation logs. Plots shall identify units, recording identity and whether a reference curve is measured, annotated or controlled.

Non-functional Requirements

Table 5. Testable non-functional requirements

Requirement	Proposed acceptance test
Response time	Process a supported 20-second phrase within 30 seconds on the documented demonstration machine; report median and 95th-percentile latency. This is a development target.
Reliability	All corrupt, empty and invalid-sample test files return a controlled status; no failure overwrites the original.
Preservation	Retain duration to within one target-rate sample after resampling and preserve timestamp origin; log every edit and crossfade.
Privacy and security	Local prototype processing, restricted research storage, coded identifiers and an explicit consent/deletion procedure; no automatic research reuse.
Usability and accessibility	Essential actions are keyboard-operable, controls have text labels, feedback does not rely on colour, and audio findings have text explanations. Verify by checklist and pilot tasks.
Reproducibility	Each experiment records source split, seed, configuration, checkpoint, policy and software versions.
Scalability	One bounded phrase per local request in the prototype; concurrent-service performance is not a claimed deliverable.

6. COMMERCIALIZATION PLAN

The proposed solution will be commercialized as a web-based Software-as-a-Service (SaaS) platform supported by a freemium business model. The trained Harmony-oriented melody understanding model and the affiliated documentation will be released as open source for further research and educational purposes. The web platform will provide hosted AI-powered harmonization services. Revenue generated through the premium subscriptions and the advertisements in the websites will be used to cover the cloud computing, hosting, and maintenance costs while ensuring the long-term sustainability of the platform.

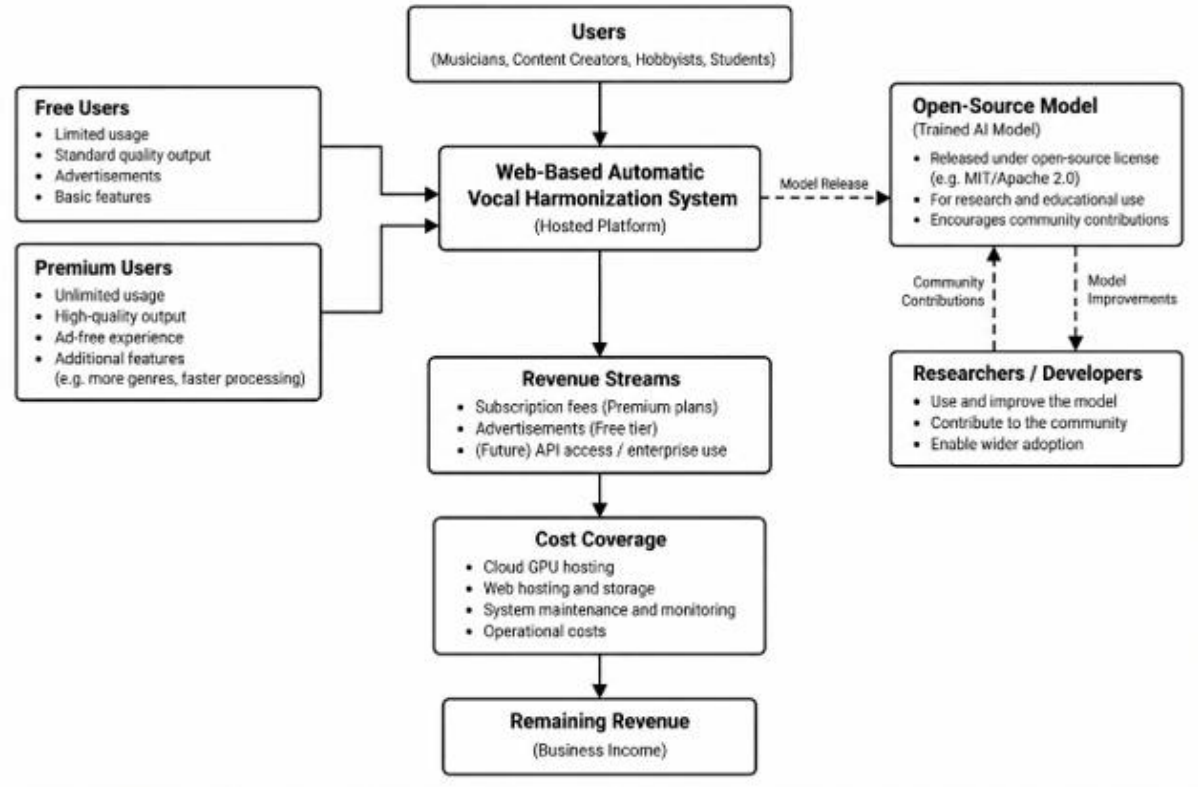


Figure 3. Proposed Business Model

6.1 Target Market & Customer Persona

The proposed system targets individuals in the musical industry who require quick and accessible way of harmonizing melodies without advanced expertise of music theory. The target market includes,

- Independent Musicians
- Music Producers and Composers
- Content Creators
- Music Students and Educators
- Music Enthusiasts & Hobbyists

6.1.1 Example Customer Persona

Table 6. Example Customer Persona

Attribute	Description
Name	Process a supported 20-second phrase within 30 seconds on the documented demonstration machine; report median and 95th-percentile latency. This is a development target.
Age	All corrupt, empty and invalid-sample test files return a controlled status; no failure overwrites the original.
Occupation	Retain duration to within one target-rate sample after resampling and preserve timestamp origin; log every edit and crossfade.
Needs	Local prototype processing, restricted research storage, coded identifiers and an explicit consent/deletion procedure; no automatic research reuse.
Expected Benefit	Fast, AI-assisted harmonization through an easy-to-use web application
Challenges	Essential actions are keyboard-operable, controls have text labels, feedback does not rely on colour, and audio findings have text explanations. Verify by checklist and pilot tasks.

6.2 Value Proposition

The proposed system unlike existing current solutions enables users to generate genre-aware harmonized musical arrangements directly from their raw vocal recordings through a simple hosted web interface. Unlike traditional music production and composition this does not require domain expertise for the user. User doesn't need extensive knowledge in music theory or production concepts or how the underlying deep learning architecture works.

In addition, the, the complete deep learning hierarchy will be released as an open-source contribution, encouraging transparency, reproducibility and enabling further research while the hosted platform provides convenient production ready service.

6.3 Revenue Model

The proposed platform will adopt a. Freemium Software-as-a-Service (SaaS) business model. Users can subscribe using monthly or annual subscription plans.

6.3.1 Free Tier and Premium Tier

Table 7. Free vs Premium Tier Features

Feature	Free Tier	Premium Tier
Daily Generations	Limited	Unlimited
Audio Quality	Standard quality	High-quality output (WAV supported)
Advertisements	Displayed	Ad-free experience
Processing Priority	Standard queue	Priority GPU queue
Genre Selection	Basic genres	Full genre library

Download Options	Standard quality downloads	High-quality downloads
Commercial Usage Rights	Not included	Included
Subscription Fee	Free	Monthly or annual subscription

Revenue will primarily be generated through **premium subscriptions & advertisements displayed to free users**. Future revenue opportunities may include **API licensing** and **enterprise subscriptions for third-party music applications**.

6.4 Cost Estimation

The primary operational costs associated with the proposed platform are summarized below.

Table 8. Operational Costs of Proposed System

Cost Category	Purpose
Cloud GPU Computing	Deep learning model inference
Web Hosting	Hosting the web application
Cloud Storage	Audio uploads and generated outputs
Database Services	User accounts and application data
Domain Registration	Website domain
System Maintenance	Updates, monitoring, and bug fixes

Revenue generated through premium subscriptions and advertisements is expected to cover these operational costs, with remaining revenue contributing to future development and platform maintenance.

6.5 Competitive Advantage

The proposed system offers several advantages compared to existing AI-assisted music generation solutions.

Table 9. Comparison with existing platforms

Existing Platforms	Proposed System
Primarily closed-source AI models	Open-source Harmony-Oriented Melody Understanding Model
Limited transparency	Research-driven and reproducible implementation
Focus on complete music generation	Dedicated vocal melody understanding followed by harmonization
Commercial-only deployment	Hybrid open-source model with hosted SaaS platform
Limited research accessibility	Model available for education and further research

By combining an open-source AI model with a commercially hosted web platform, the proposed solution promoted both innovation and long-term sustainability.

6.6 Intellectual Property Considerations

The Harmony-Oriented Melody Understanding Model developed through this research will be released under an opensource license. This supports academic research, transparency and provides a valid community contribution. The source code of the hosted web application, User Interface , Platform branding, and deployment infrastructure will remain the intellectual property rights of the research and development team.

Users will retain ownership of the musical content that they generate using the platform. The system acts solely as an AI-assisted tool for music generation and do not claim ownership for inputs that users enter or the musical outputs they generate This approach ensures that creators have full rights over their generated music while the platform provides its services under clearly defined terms of use.

7. BUDGET AND JUSTIFICATION

Table 10 presents proposed spending ceilings in Sri Lankan rupees, not supplier quotations or confirmed institutional resources. Purchases and service use will require approval and current quotations. The plan assumes access to an existing computer and a single-voice recording device; if either is unavailable, the budget and schedule must be revised before recruitment. No hardware purchase is silently assumed to be funded.

Table 10. Proposed component budget in LKR

Category	Estimated Cost (LKR)	Justification
Hardware	0	Existing personal computers will be used.
Software Licenses	0	Open-source software and educational licenses will be used.
Cloud Services	45,000	GPU inference, cloud hosting, storage, and deployment.
Human Effort	To be Decided (TBD)	Estimated development and research effort.
Miscellaneous	10,000	Domain registration and contingency expenses.
Total Estimated Budget	**55,000**	

8. WORK BREAKDOWN STRUCTURE (WBS)

Table 11 allocates the individual component workload. The hour figures are planning estimates and will be revised using pilot effort. Deliverables are arranged so that data quality and baseline evidence precede the full model comparison. Weekly progress will be tracked against completed outputs rather than an inflated count of generated audio files.

Table 11. Individual work packages and effort estimates

Work package	Deliverable	Weeks	Hours
W1 Literature and gap analysis	Prior-work matrix and bounded question	1-6	50
W2 Design, protocol and ethics	Experimental plan and interface contract	3-8	50
W3 Data and intervention preparation	Permissions, pilot, partitions and audited pairs	7-18	120
W4 Models and controlled experiments	Three trainable models and controlled comparisons	15-28	140
W5 Prototype and preparation	Capture, controller, rendering and feedback	9-28	90
W6 Integration	Accepted-package and downstream contract tests	23-32	50
W7 Evaluation and analysis	Held-out metrics, listening and failure analysis	29-38	100
W8 Documentation and assessment	Reproducible report and demonstration	1-40	40
Total	Individual component	40 weeks	640

9. GANTT CHART

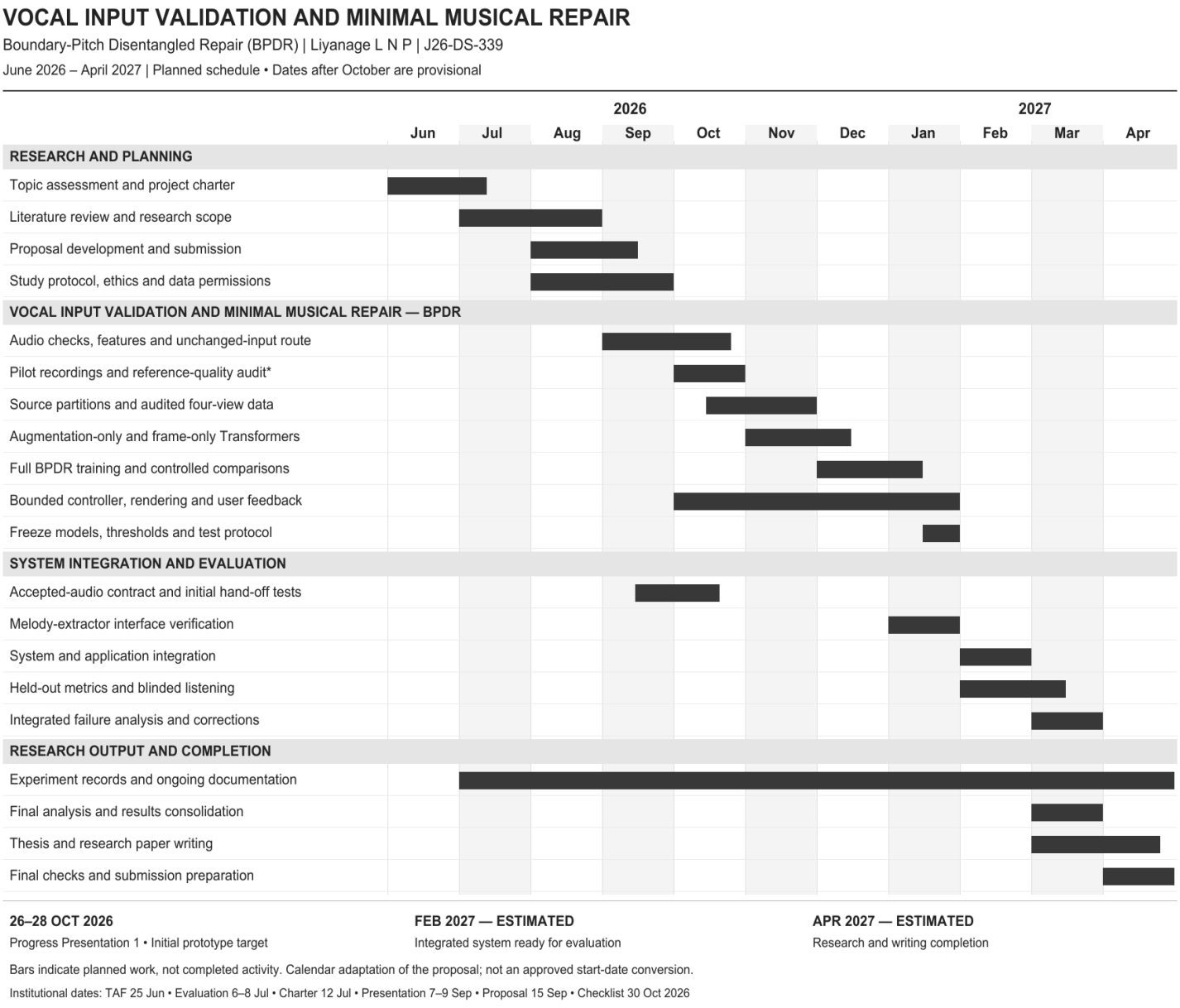


Figure 3. Proposed project schedule

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11. APPENDICES

Appendix 1 - Proposed Research Instruments and Experimental Records

Requirements elicitation and listening protocol

The following instruments are proposed and have not yet been administered. The recording session will explain the supported phrase length and distinguish participation consent, research reuse and optional demonstration permission. A participant may provide a guide melody rendition and a freely chosen hum. The researcher will record device conditions, prior singing experience as self-reported, recording attempts and participant comments without inferring demographic characteristics from the voice.

The elicitation discussion will ask: What were you trying to communicate? Which feedback helped you record another take? Can you identify what changed between the two versions? Does the prepared version still represent your musical idea? Would you prefer the unchanged version, the prepared version or another attempt, and why? No answer will be presumed in advance.

For blinded assessment, listeners will receive randomized paired clips at comparable playback levels. They will separately judge preservation of the tune, expression and audible processing artefacts using a pilot-tested five-point scale with verbal anchors. Method labels and controlled detuning details will remain hidden until the judgement is recorded. The final rubric and recruitment count will be fixed before held-out assessment.

Experimental record and integrity checks

Each source record will contain a coded singer identifier, phrase identifier, permission record, original checksum, recording conditions, reference-suitability judgement, boundary annotations, partition and exclusion reason where relevant. Each transformed descendant will additionally record intervention type, cents curve, interval support, renderer configuration, identity-control relationship and realized-shift audit.

Each evaluation record will identify one of the five conditions in Table 2, the source and any accepted version, status, model and seed where applicable, controller policy, software configuration, latency and observed failure. Controlled reference plots will be labelled separately from measured curves. No graph in this proposal represents a trained BPDR result. Experimental plots will be generated from actual saved runs during evaluation.

Table 12. Proposed failure and preservation tests

Case	Expected behaviour to test
Clear acceptable hum	Musical no-change acceptance without unnecessary tuning.
Small local detuning	Supported local repair, with preservation elsewhere.
Identical audio with false split or merge	Stable correction relative to the nominal boundary view.
Vibrato, slide or uncertain transition	Preserve expression or withhold the edit.
Mild steady noise with an eligible noise interval	Compare conservative attenuation against unchanged input.
Severe clipping or unrecoverable melodic evidence	Explain the observation and request another take.
Invalid file or rendering failure	Return controlled status or rollback; never overwrite the original.

Ethics and feasibility safeguards

Ethics approval and informed consent will precede recruitment. Coded audio will remain in restricted research storage for the approved retention period. Withdrawal and deletion procedures will be explained before recording. Public release of clips will require separate permission. The study will avoid claims about a participant's musical ability based only on model uncertainty. The main scientific risk is that BPDR adds no benefit beyond augmentation or removal of boundary input. The main data risk is insufficient trustworthy reference material. Rendering artefacts and limited benefit to the final extractor are additional risks. Matched controls, a pilot target audit, local rollback and independent listening are the corresponding mitigations. These measures support a bounded, implementable study; they do not predetermine positive findings.

Appendix 2 - Mandatory Proposal Evidence and Decision Log

The following entries reconstruct the development of the research problem and proposed system from June to 15 September 2026. They record experience-based observations, literature findings, discussions, evaluation feedback and resulting design decisions. Except for 6 July and 2 September, dates should be verified against meeting notes. Entries do not represent completed experiments or formal supervisor approval unless explicitly stated.

Table 13. Proposal evidence and decision log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
16 Jun 2026	Uploaded vocal audio may be unsuitable for processing	Vocal-quality risks; annotation ambiguity [2]	Accept all; repair all; validate first	Validate input before melody extraction	Component role discussed
22 Jun 2026	Automatic repair may alter the user's melody	Expressive pitch-correction research [1], [4], [5]	Full correction; no correction; bounded repair	Use preservation-first, limited repair	Discussed as ownership/safety issue
6 Jul 2026	Component boundaries required clarification	Workflow and evaluation feedback	End-to-end model; loosely linked stages; explicit components	Keep four components with clear hand-offs	Reviewed at 6 July evaluation
11 Sep 2026	Boundary errors may affect pitch repair	BERT-APC segmentation sensitivity [1]; vocal annotation evidence [2]	Boundary-free model; ordinary augmentation	Test false split/merge robustness explicitly	Pending review
11 Sep 2026	General consistency learning is existing work	SPICE, PESTO and Mean Teacher [7], [8], [10]	Claim generic consistency novelty	Claim only the specific BPDR formulation	Pending review
11 Sep 2026	Boundary and pitch changes need separate testing	Contextual correction and transformation learning [1], [8], [10]	Single augmentation; separate unrelated tests	Use paired four-view BPDR training design	Pending review

Appendix 3 - AI Use Disclosure

Generative AI tools were used during the preparation and development of this proposal as supporting tools. AI assistance was used for research organization, explanation, drafting, revision, technical discussion, diagram development and document preparation. AI-generated suggestions were not treated as experimental evidence or independently verified research findings. No AI-generated performance values, participant responses, signatures, ethics approvals or experimental results are presented as observed results.

Table 14. AI assistance and verification record

AI Tool	Purpose of Use	What Was Generated / Assisted	How Output Was Verified	Final Student Contribution
ChatGPT (OpenAI)	Literature review and research synthesis	Summaries and comparisons of relevant literature, identification of related concepts, organization of prior work and discussion of the BPDR research gap	Claims were checked against the original research papers, official documentation and cited sources before inclusion	Student selected the relevant literature, determined its relevance to the research problem and remains responsible for the final interpretation and citations
ChatGPT (OpenAI)	Research-methodology development	Discussion and refinement of the BPDR four-view design, boundary interventions, pitch interventions, baselines, ablations, evaluation metrics and validation strategy	Methodological suggestions were compared with the research objectives, primary literature and feasibility constraints; unsupported suggestions were revised or excluded	Student made the final methodological decisions and is responsible for implementing, testing and defending the proposed method
ChatGPT (OpenAI)	Technical explanation and reasoning	Assistance in explaining pitch correction, note boundaries, Transformer architecture, loss functions, controller logic, data flow and experimental comparisons	Technical statements were checked against cited publications, equations and official software documentation	Student reviewed the technical reasoning and determined the final system architecture and technical choices
ChatGPT (OpenAI)	Academic drafting and revision	Assistance with sentence construction, paragraph organization, clarity, conciseness, terminology and restructuring of proposal sections	Drafted material was manually reviewed against the intended research meaning and supporting references	Student edited and approved the final wording and remains responsible for originality, accuracy and consistency
ChatGPT	Critical review	Identification of	Recommendations were	Student decided which

(OpenAI)	of proposal content	possible weaknesses, unclear claims, unsupported statements, alternative explanations, methodological risks and areas requiring stronger validation	checked against the proposal objectives and supporting literature rather than accepted automatically	recommendations to adopt and made the final research decisions
ChatGPT (OpenAI)	Diagram and workflow assistance	Assistance in planning and refining system architecture diagrams, BPDR workflow diagrams, training-flow diagrams and status-routing logic	Diagrams were checked against the written methodology, system architecture and component responsibility boundaries	Student reviewed and edited the diagrams to ensure that they accurately represent the proposed system
ChatGPT (OpenAI)	Tables and document organization	Assistance in organizing comparison tables, requirements, evaluation criteria, work packages, schedules and appendix material	Entries were compared with the corresponding proposal sections and source information	Student checked and approved the final organization and content
ChatGPT (OpenAI)	Reference and documentation support	Assistance in organizing references, identifying information requiring verification and locating relevant documentation for software and methods	Bibliographic details and technical claims were checked using the original papers or official documentation	Student remains responsible for reference validity, citation accuracy and appropriate use of sources
ChatGPT (OpenAI)	Formatting and presentation support	Assistance with proposal structure, headings, table formatting, figure captions and presentation consistency	Final document was manually reviewed against the required proposal structure and submission requirements	Student completed the final review and accepts responsibility for the submitted document